Do Attention-Derived Hopfield Energies Contain Hallucination Signal?

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Abstract

Large language models can produce answers that are unsupported by the provided context or false with respect to common knowledge. This project studies whether such failures are visible in the internal attention geometry of a frozen model. We extract query/key vectors after rotary position embedding and compute modern-Hopfield-style energy and prompt-attention summaries over response tokens. On a small SMILES/SQuAD-derived dataset, these features improve a response-likelihood baseline from 0.689 to 0.723 AUROC. On RAGTruth, the same feature family transfers across three small models: Qwen2.5-0.5B reaches 0.707 ± 0.005 AUROC, SmolLM2-360M reaches 0.705 ± 0.003, and Gemma-3-1B reaches 0.679 ± 0.003. A TruthfulQA stress test reaches 0.706 ± 0.003 AUROC, but also shows that scalar energy is not a universal monotonic hallucination score. The main conclusion is therefore limited but positive: Q/K energy and attention-derived summaries contain transferable hallucination/factuality signal, while their interpretation depends on the hallucination type and model architecture.

1 Introduction

Hallucination detection is often studied through outputs: whether a generated answer agrees with a source passage, retrieved evidence, or external facts. In this project, I ask a narrower diagnostic question: can a frozen language model's own attention geometry provide a useful signal for detecting hallucinated or false answers?

The motivation comes from the connection between attention and modern Hopfield networks. Modern Hopfield networks implement associative retrieval by comparing a query to stored patterns with a softmax-like rule [Ramsauer et al., 2021]. Transformer attention has the same query/key/value structure [Vaswani et al., 2017]. If a response token is weakly supported by the prompt or by previous tokens, then its query/key energy or attention-grounding statistics may differ from a well-supported token. This does not imply that energy alone should be a perfect detector. It suggests that attention-derived features may contain information complementary to ordinary likelihood.

The study focuses on two hallucination settings. The first is context-grounded hallucination: an answer is unsupported by or contradictory to a supplied passage. This is tested on the original SMILES/SQuAD-derived dataset and on RAGTruth [Wu et al., 2024]. The second is open-domain false-answer detection: an answer reflects a common misconception or false belief, tested with TruthfulQA [Lin et al., 2022]. These settings are intentionally separated because they need not have the same internal signature.

The contributions are:

- I implement a train-free feature extractor that computes Q/K Hopfield energy and prompt-attention summaries from selected attention layers of frozen causal language models.
- I compare these features against response negative log-likelihood and response length on three datasets and three small model families.

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- I show that the best combined probe is stable over five cross-validation seeds and transfers from Qwen to SmolLM2 and Gemma on RAGTruth.
- I show an important limitation: scalar energy direction is not universal. It is meaningful on RAGTruth but changes on TruthfulQA, so multifeature probes are safer than a single energy score.

2 Background and hypothesis

Hallucination types. The project distinguishes context-grounded hallucination from open-domain falsehood. In the context-grounded setting, the model answer is judged against an explicit passage or retrieved evidence. In the open-domain setting, the answer is judged against external factual knowledge or common misconceptions. This distinction follows the broader separation between faithfulness to a source and factuality with respect to the world [Ji et al., 2023, Maynez et al., 2020].

Modern Hopfield energy. For a response-token query q_t and previous-key matrix $K_{\leq t}$, I compute the attention-score log-sum-exp energy

$$E_t = -\sqrt{d_h} \log \sum_{i \leq t} \exp\left(\frac{q_t^\top k_i}{\sqrt{d_h}}\right), \tag{1}$$

where d_h is the head dimension. This is not a new attention mechanism and it does not change the model output. It is a diagnostic computed from the frozen model's Q/K states after rotary position embedding.

Hypothesis. The working hypothesis is that hallucinated or false responses should be partly visible in attention geometry. In a context-grounded setting, unsupported answers may have weaker associative support from the prompt and therefore higher, less stable, or differently distributed Q/K energy. In an open-domain setting, the signal may still exist, but it may not have the same scalar direction because there is no explicit evidence passage to ground the response.

3 Method

Each example is represented as a prompt, a response, and a binary label where label 1 means hallucinated or false. For each prompt-response pair, I run the frozen language model on the concatenated text and score only the final response tokens. The default maximum is 32 response tokens and 512 total tokens.

For selected layers, the extractor registers a forward pre-hook on self-attention and reconstructs query/key tensors after rotary position embedding. For every selected layer and head, it computes:

- Hopfield energy statistics over response tokens: mean, standard deviation, maximum, and final-token energy;
- prompt-attention statistics: mean prompt attention mass, final-token prompt attention mass, mean maximum prompt attention, and mean prompt-attention entropy.

The final feature sets either average these quantities across selected layers or keep them as per-layer features.

The main classifier is intentionally simple: a standardized logistic regression with balanced class weights. The goal is not to maximize leaderboard performance, but to test whether the feature family contains stable signal. I compare the following baselines and probes:

- response length and response negative log-likelihood (NLL);
- NLL + length logistic regression;
- Hopfield-energy-only and attention-only probes;
- combined NLL + length + Hopfield/attention probes.

All learned probes are evaluated with five-fold stratified cross-validation. The final cached evaluation repeats this over seeds $13, 21, 42, 87, 101$ and reports mean and standard deviation. For context-grounded datasets, I also run a context-grouped split when repeated contexts are available.

Table 1: Cached multi-seed evaluation. Values are AUROC mean $\pm$ standard deviation over five cross-validation seeds.

Dataset	Model	NLL	NLL+len	Energy probe	Attention probe	Best combined
SMILES/SQuAD	Qwen2.5-0.5B	0.664 ± 0.000	0.689 ± 0.004	0.694 ± 0.006	0.673 ± 0.006	$\mathbf{0.723 \pm 0.010}$
RAGTruth	Qwen2.5-0.5B	0.550 ± 0.000	0.652 ± 0.007	0.703 ± 0.002	0.680 ± 0.003	$\mathbf{0.707 \pm 0.005}$
RAGTruth	SmolLM2-360M	0.537 ± 0.000	0.642 ± 0.005	0.700 ± 0.001	0.680 ± 0.003	$\mathbf{0.705 \pm 0.003}$
RAGTruth	Gemma-3-1B	0.538 ± 0.000	0.647 ± 0.003	0.672 ± 0.002	0.654 ± 0.005	$\mathbf{0.679 \pm 0.003}$
TruthfulQA	Qwen2.5-0.5B	0.400 ± 0.000	0.578 ± 0.004	0.651 ± 0.004	0.666 ± 0.006	$\mathbf{0.706 \pm 0.003}$

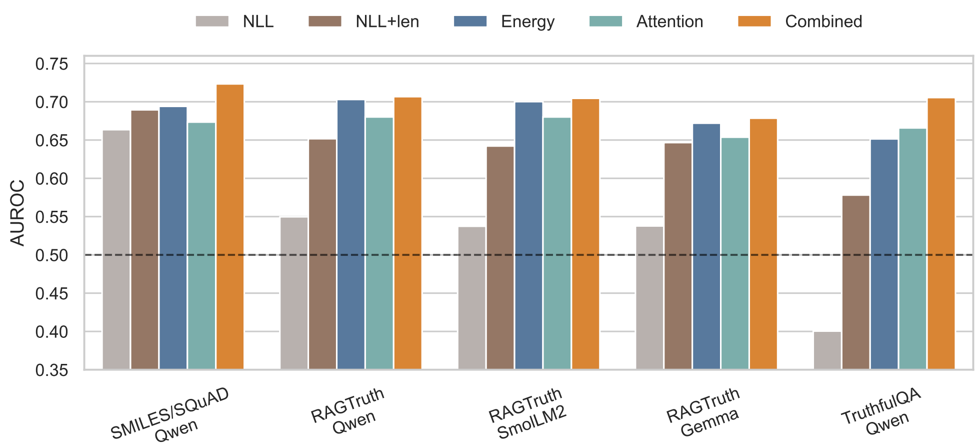

Figure 1: Feature-family comparison across datasets and models. The combined probe consistently improves over response NLL; energy and attention features are most useful on RAGTruth.

4 Experimental setup

Datasets. The first dataset is the SMILES hallucination-detection application dataset, a 689-example SQuAD-2.0-derived benchmark distributed with the public task template [Rajpurkar et al., 2018, SMILES-2026 Project Authors, 2026]. It contains 206 truthful and 483 hallucinated context-grounded QA examples where generated answers may be unsupported by the supplied passage.

RAGTruth [Wu et al., 2024] is used as a larger context-grounded transfer benchmark. I convert the processed test split into 2700 prompt-response-label rows: 1757 truthful and 943 hallucinated. I mark an example as hallucinated when the processed label contains evident conflict or baseless information.

TruthfulQA [Lin et al., 2022] is used as an open-domain stress test. I convert the generation split into paired rows: one correct answer with label 0 and one incorrect answer with label 1 for each question, giving 1634 examples. This is not the same task as generated-answer hallucination detection, so I treat it as a test of false-answer separability rather than direct deployment performance.

Models. The main model is Qwen2.5-0.5B [Team, 2025]. For model-transfer checks on RAGTruth, I also evaluate SmolLM2-360M-Instruct [Allal et al., 2025] and Gemma-3-1B-it [Gemma Team et al., 2025]. For Qwen, the selected layers are 12, 16, and 20. For SmolLM2 and Gemma, the selected layers are 6, 12, and 20.

5 Results

Table 1 gives the main result. Across all settings, the best combined Hopfield/attention probe improves over NLL and NLL+length. On the original SMILES/SQuAD dataset, the improvement over NLL+length is about 3.4 AUROC points. On RAGTruth, the improvement is larger: Qwen improves from 0.652 to 0.707, and SmolLM2 improves from 0.642 to 0.705.

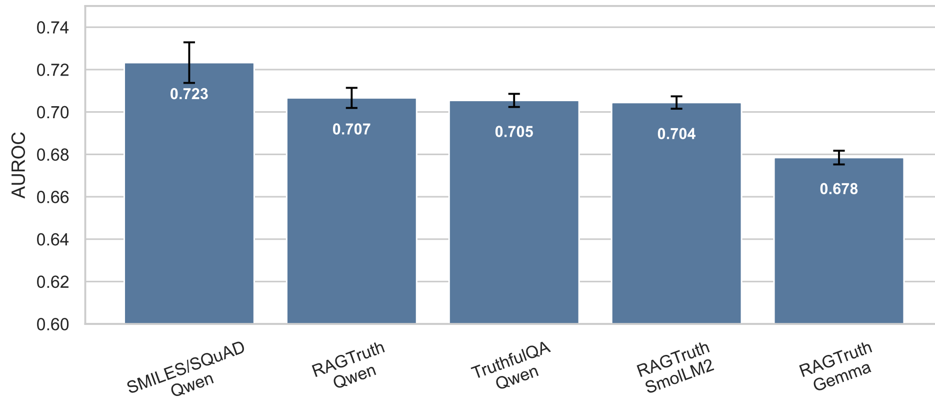

Figure 2: Best combined AUROC across dataset/model settings. Error bars show standard deviation over five cross-validation seeds.

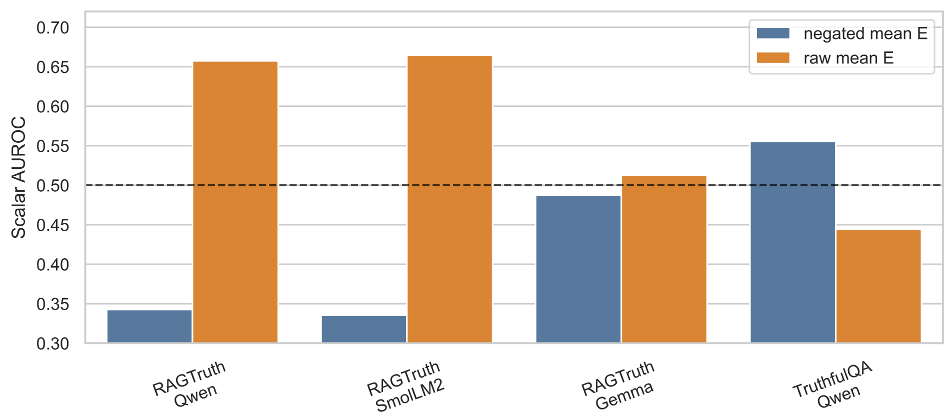

Figure 3: Scalar mean-energy AUROC differs by dataset. It is useful on RAGTruth for Qwen and SmolLM2, weak on Gemma, and reverses direction on TruthfulQA.

5.1 Context-grounded hallucination

The strongest context-grounded result is RAGTruth. In this setting, scalar response NLL is weak: 0.550 AUROC for Qwen and 0.537 for SmolLM2. By contrast, per-layer Hopfield energy reaches 0.703 and 0.700 respectively. This suggests that the Q/K geometry captures a support signal that ordinary likelihood misses.

The result is not purely Qwen-specific. SmolLM2-360M gives almost the same RAGTruth AUROC as Qwen2.5-0.5B. Gemma-3-1B is weaker, reaching 0.679 with the best combined probe, but it still improves over NLL+length. This weakens any claim that the method is architecture-invariant, but supports the more modest claim that the signal is not limited to a single model family.

5.2 Open-domain false answers

TruthfulQA gives a different picture. The best combined probe still reaches 0.706 AUROC, but the scalar features behave differently. Raw response NLL has 0.400 AUROC, and raw mean Hopfield energy has 0.444 AUROC in the single-run scalar evaluation. Negating the energy score improves the scalar direction, while multifeature probes remain useful.

This result changes the interpretation of the method. A simple rule such as "higher energy means hallucination" is too strong. The safer conclusion is that attention-derived Q/K features contain factuality-relevant signal, but the sign and useful aggregation depend on the task.

6 Discussion

The results support the main hypothesis in a limited form. Attention-derived Hopfield energy and prompt-grounding summaries contain hallucination signal beyond response likelihood. The strongest evidence is RAGTruth, where energy-only per-layer probes outperform NLL+length on Qwen and SmolLM2. The multi-seed standard deviations are small, so the effect is not just a lucky split.

At the same time, the study argues against an overly simple energy story. Energy is not a universal scalar detector. On TruthfulQA, the scalar direction reverses, and Gemma shows weaker scalar energy on RAGTruth than Qwen or SmolLM2. This points to a more cautious interpretation: Q/K geometry is a useful representation for probing, not a standalone law of hallucination.

I also interpret the method as diagnostic rather than deployable. The classifier uses hidden attention internals and requires access to model weights. It is therefore useful for understanding model behavior and designing detectors, but it is not directly applicable to closed black-box systems.

7 Limitations

The datasets are not identical in meaning. SMILES/SQuAD and RAGTruth evaluate context-grounded support, while TruthfulQA evaluates correct versus incorrect answer candidates. These should not be merged into one universal hallucination benchmark.

The model set is still small. Qwen, SmolLM2, and Gemma provide useful architectural diversity, but the results may change for larger models, different layer choices, or models with substantially different attention implementations.

The classifier is a probe trained on labels from the same dataset. Although cross-validation and context-grouped splits reduce leakage concerns, the reported AUROC should not be interpreted as out-of-domain deployment performance.

The method requires access to attention internals. It is not compatible with black-box APIs unless the provider exposes equivalent intermediate states. It also adds extraction cost because Q/K tensors must be processed for selected layers and response tokens.

Finally, the current features average over heads. This may hide a more localized mechanism. A future analysis should study per-head contributions and layer sensitivity rather than only selected-layer summaries.

8 Conclusion

This project tested whether attention-derived Hopfield energy and prompt-attention summaries contain hallucination signal. The answer is positive but qualified. Across three datasets and three small model families, simple probes over Q/K-derived features consistently improve over response likelihood baselines. The strongest RAGTruth results transfer from Qwen2.5-0.5B to SmolLM2-360M, and a weaker but positive result is observed for Gemma-3-1B.

The main scientific conclusion is not that Hopfield energy is a universal hallucination score. It is that Q/K attention geometry provides a useful representation for factuality probing, while the scalar energy direction depends on the hallucination type and architecture. This makes the result more limited, but also more informative: internal attention geometry appears to encode support and factuality signals, but these signals need careful calibration rather than a single fixed threshold.

References

Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, Gabriel Martín Blázquez, Guilherme Penedo, Lewis Tunstall, Andrés Marafioti, Hynek Kydlíček, Agustín Piqueres Lajarín, Vaibhav Srivastav, Joshua Lochner, Caleb Fahlgren, Xuan-Son Nguyen, Clémentine Fourrier, Ben Burtenshaw, Hugo Larcher, Haojun Zhao, Cyril Zakka, Mathieu Morlon, Colin Raffel, Leandro von Werra, and Thomas Wolf. Smollm2: When smol goes big – data-centric training of a small language model, 2025.

Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, et al. Gemma 3 technical report, 2025.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), 2023.

Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, 2022.

Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. On faithfulness and factuality in abstractive summarization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020.

Pranav Rajpurkar, Robin Jia, and Percy Liang. Know what you don't know: Unanswerable questions for squad. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 784–789, 2018. doi: 10.18653/v1/P18-2124.

Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021.

SMILES-2026 Project Authors. SMILES-2026 Hallucination Detection. `https://github.com/ahdr3w/SMILES-2026-Hallucination-Detection`, 2026. Public application project repository and dataset template.

Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

Yuanhao Wu, Juno Zhu, Siliang Xu, Kashun Shum, Cheng Niu, Randy Zhong, Juntong Song, and Tong Zhang. Ragtruth: A hallucination corpus for developing trustworthy retrieval-augmented language models, 2024.